ICFCR 2026

International Conference on Futuristic Technologies in Control Systems & Renewable Energy 2026

Sept 23-24, 2026 @ MESCE Kuttippuram

Paper ID : 105

Paper Title : Non-Invasive Leakage Severity Estimation in Water Distribution Networks using Wavelet Scattering Transforms: A Comparative Sensor Study

Authors : Shiju Kumar P S • Govind Jee (Presenter) • Raj Kishor

Affiliation : Department of Computing Technologies, SRM Institute of Science and Technology

Why Leakage Severity Matters

From binary alarm to actionable severity estimate — motivating hydraulic severity classification

The Water Loss Problem

Non-Revenue Water (NRW) can consume **up to half** of municipal supply.

Binary leak detection answers only:

'Is there a leak?'

Operational Gap:

'Which leak deserves priority? What is the severity?'

Hydraulic severity classification bridges this gap:
mapping non-invasive pipe-wall signals to discrete flow-magnitude states for maintenance prioritization.

Research approach: WST-based classification on a public experimental benchmark — computationally lightweight, no hydraulic model calibration required.

The Research Question

Can **WST features** from non-invasive pipe-wall vibration distinguish **three hydraulic severity states** without fluid contact?

No Demand (ND)
Hydraulic control state

Low Flow • 0.18 L/s
Lower-severity — maintenance priority

High Flow • 0.47 L/s
Higher-severity — maintenance priority

*… and how does **external vibration** compare against **invasive in-pipe sensors** under identical benchmark conditions?*

Binary Alarm 'Present / Absent' — **Severity Estimation** 'ND / 0.18 / 0.47 L/s' — **Maintenance Prioritization** 'Lower / Higher priority'

Research Gap, Prior Work & Contribution

Positioning this study within the leak detection and flow-severity estimation landscape

Binary Detection	Leak-Type Classification	Regression Sizing	Severity Estimation
"Leak present?"	"Crack / Gasket / Orifice?"	"Continuous flow rate?"	"ND / Low / High?" ▶ THIS STUDY

Task Focus	Representative Prior Work	Output Provided	Gap Addressed
Binary Leak Detection	ML on FFT / statistical features (Ismail et al., Mashford et al.)	Presence / absence only (no flow rate)	No magnitude information
Leak-Type Classification	WST + CNN (Martínez-Ríos et al.) multi-sensor fusion	Crack / gasket / orifice type (no flow magnitude)	Type ≠ severity
Severity Estimation ▶ This Study	WST + RBF-SVM on non-invasive pipe vibration	Three-state triage: ND / 0.18 L/s / 0.47 L/s	Non-invasive; no model needed

Note: *Cross-study accuracy comparison must be interpreted cautiously — datasets, pipe materials, topologies, and flow conditions differ.*

Key Contribution: Comparative evaluation of three sensor modalities for hydraulic flow-severity classification using WST + SVM under benchmark conditions.

Contribution 1

Sensor Modality Comparison

3 physical sensing domains (Vibration / Pressure / Acoustic) compared under identical benchmark conditions

Contribution 2

WST + RBF-SVM Framework

5-fold stratified CV
Accelerometer: 99.0% ± 1.2%
Pressure: 93.7% ± 1.9%

Contribution 3

Non-Invasive Vibration Evidence

External accelerometer reliably indicates hydraulic flow severity under benchmark testbed conditions

Benchmark Dataset & Provenance

Public testbed dataset (Aghashahi et al., 2022) — benchmark structure vs. study scope

Aghashahi, Sela & Banks (2022) — *"Dataset of Leak Simulations in Experimental Testbed Water Distribution System"*

Mendeley Data · DOI: 10.17632/tbrnp6vrnj.1 · Public Experimental Benchmark Dataset

Public Benchmark Dataset Contains

Experimental Infrastructure (Aghashahi et al.)

Fabricated & recorded by dataset authors (HDPE 25 mm pipe)

Network Topologies

Branched · Looped

Simulated Leak Conditions

Circumferential Crack · Gasket Leak
Longitudinal Crack · Orifice Leak · No-leak

Hydraulic States Recorded

No Demand (ND) · 0.18 L/s · 0.47 L/s · Transient

Sensor Modalities

Accelerometer · Dynamic Pressure · Hydrophone

2 channels per modality · 30 s continuous recordings

Study Selection

Severity Evaluation

Scope Selected for This Study

Target Research Problem

Hydraulic flow-severity classification (ND vs. Low vs. High)

Network Topology Selected

Branched pipe network (standard distribution branch scenario)

Leak Condition Evaluated

Orifice leak (primary localized leak simulation)

3 Hydraulic Severity States

No Demand (ND) · Low (0.18 L/s) · High (0.47 L/s)

3 Sensor Modalities Evaluated

External Accelerometer (pipe-wall vibration)
Dynamic Pressure Sensor (in-pipe fluid pressure)
Hydrophone (in-fluid acoustic emission)

Public benchmark dataset: *Aghashahi, Sela & Banks (2022), Mendeley Data. The study uses measurements acquired on the published experimental testbed described in the referenced literature. Our work develops and evaluates the WST–SVM severity-classification framework on the selected benchmark data.*

Study Design & Sensor Modalities

Three sensor modalities — identical benchmark conditions — different physical sensing domains

Common Benchmark Context: 25 mm HDPE Branched Network · 30 s Continuous Recordings · ND · 0.18 L/s · 0.47 L/s

Accelerometer
EXTERNAL · NON-INVASIVE

Signal domain:
Pipe-wall structural vibration
Sampling rate:
51.2 kHz
Sensor model:
PCB 333B50
Installation:
Clamp-on (external surface)

Invasive?
NO — no pipe penetration required

Dynamic Pressure
IN-PIPE · INVASIVE

Signal domain:
Fluid pressure transients
Sampling rate:
51.2 kHz
Sensor model:
PCB 102B16
Installation:
In-pipe (direct fluid contact)

Invasive?
YES — requires pipe penetration

Hydrophone
IN-FLUID · INVASIVE

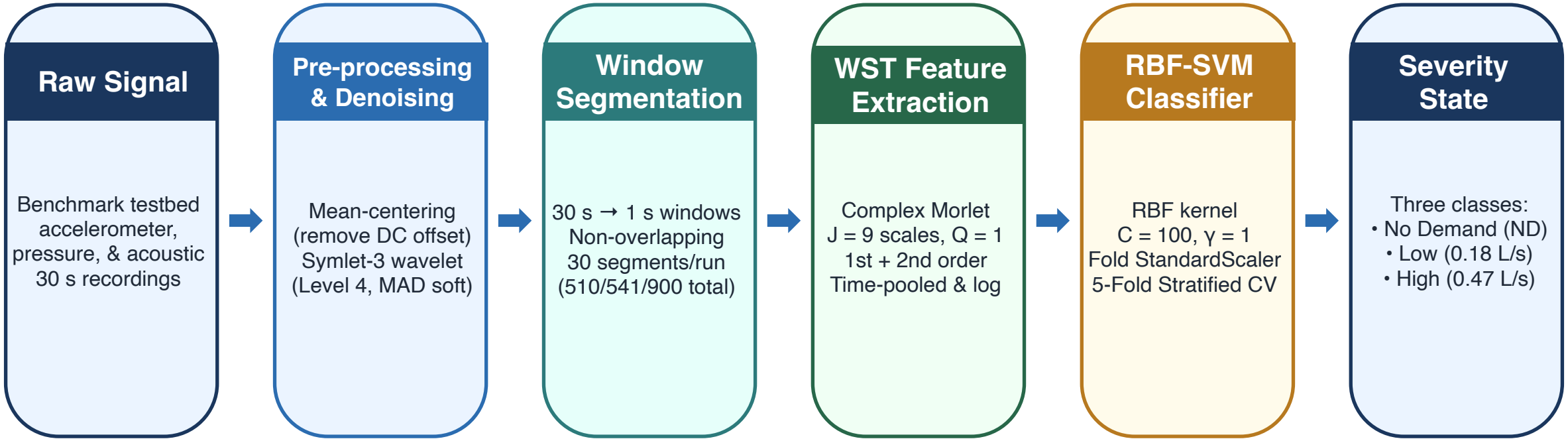
Signal domain:
Acoustic emission in fluid
Sampling rate:
8.0 kHz
Sensor model:
Aquarian H2a
Installation:
Submerged in fluid

Invasive?
YES — submerged in fluid column

Evaluation context: All three modalities recorded under identical hydraulic states on the same benchmark testbed — enabling direct cross-domain comparison of **non-invasive vibration** vs. **invasive fluid sensing** vs. **in-fluid acoustics.**

WST–SVM Framework Pipeline

Five processing stages transforming raw benchmark signals into hydraulic severity decisions



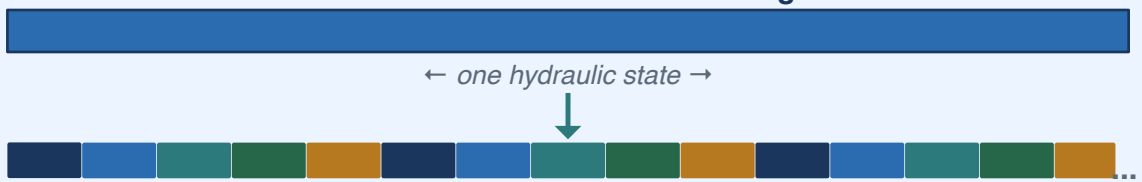
Modality-Independent Execution: External accelerometer, dynamic pressure, and hydrophone signals are each evaluated independently through this identical pipeline. No features or parameters are shared across modalities.
Fold-Isolated Feature Scaling: StandardScaler was fitted only on the training partition within each cross-validation fold. Denoising, feature extraction, and SVM training operate strictly within each fold.

Signal Preprocessing & Segmentation

From 30-second recordings to conditioned 1-second machine-learning windows

From Recording to ML Samples

30-second continuous recording



30 × 1-second non-overlapping windows (= 30 ML samples per recording)

Each 1 s segment conditioned independently prior to WST feature extraction

Signal Conditioning (per 1 s segment)

① **Mean-centering:** Remove DC offset from each 1 s window

② **Wavelet denoising:** Symlet-3 wavelet — 4-level decomposition

③ **Noise estimation:** Median Absolute Deviation (MAD) of detail coefficients

④ **Soft thresholding:** Universal threshold applied; signal reconstructed

Segment Counts: Full Dataset vs. Test Splits

510 **Accelerometer Segments**
170 per class (ND, Low, High)
Representative test split: n = 102

541 **Dynamic Pressure Segments**
541 total · in-pipe fluid sensor
Representative test split: n = 109

900 **Hydrophone Segments**
300 per class (submerged acoustic)
Representative test split: n = 180

CV & Scaling Rigor

StandardScaler: fitted on training partition only (per fold).
Non-overlapping windows: adjacent segments share no raw samples.
Note: Recording-level grouped CV is not specified in the paper.

Why Wavelet Scattering Transform?

Stable multi-scale representation for non-stationary hydraulic turbulence signals

Turbulence Representation Challenge

- **Non-Stationary Turbulence:** Turbulent flow generates transient, non-stationary structural vibrations.
- **FFT Limitation:** FFT-based representations rely on stationarity assumptions that may not hold for transient turbulence signals.
- **Simple Statistics Insufficient:** Simple summary statistics do not explicitly capture the multi-scale structure targeted by WST.
- **WST Solution:** WST uses fixed analytical filters with no learned transform weights — providing stable multi-scale decomposition.

WST Configuration

Wavelet Family: Complex Morlet analytical wavelets

Dyadic Scales: J = 9 octaves (captures lowest flow modulation frequencies)

Quality Factor: Q = 1 wavelet per octave (efficient spectral coverage)

Scattering Orders: 1st + 2nd orders included · 0th (DC mean) excluded

Time Pooling: Low-pass averaging (ϕ_2^j) — ensures deformation stability

Log Compression: log(S_WST) stabilizes dynamic range for linear margin classification

WST Signal Operation Flow

① Denoised Signal
1 s segment · x(t) conditioned

② Wavelet Convolution
$x * \psi\lambda_1$ (Complex Morlet, J=9, Q=1)

③ Modulus
$|x * \psi\lambda_1|$ (phase stripped; envelope retained)

④ 2nd-Order Cascade
$||\cdot| * \psi\lambda_2$ (captures modulation structure)

⑤ Time Averaging
$* \phi_2^j$ (low-pass pooling; shift invariance)

⑥ Log Compression
log(S_WST) (dynamic range stabilization)

⑦ Feature Vector
→ Compact representation input to RBF-SVM

Classification & Evaluation Protocol

Hyperparameter grid search & 5-fold stratified cross-validation framework

Table III: SVM Grid Search (Accelerometer)

Kernel Config	C = 1	C = 10	C = 100
Linear	88.3%	91.2%	94.5%
RBF (γ = scale)	92.1%	98.4%	**99.2%**
Polynomial (d = 3)	89.5%	93.1%	95.8%

MODEL SELECTION / GRID SEARCH (Accelerometer)

Peak Grid-Search Result: 99.2% (RBF, C = 100, γ = scale)

- Decision boundaries sharpen with large regularization parameter (C = 100).
- RBF outperforms Linear and Polynomial across all tested regularization values.
- *Note: 99.2% reflects peak grid-search tuning on the accelerometer partition, distinct from the standardized cross-sensor evaluation.*

Final Cross-Sensor Evaluation Protocol

Final Model Parameters (Standardized Baseline)

RBF Kernel · C = 100 · γ = 1

Cross-Sensor Standardization Rationale:
For the cross-sensor comparison, the final RBF-SVM parameters were fixed at C=100 and γ=1 across modalities to evaluate physical sensor discriminability rather than per-modality tuning.

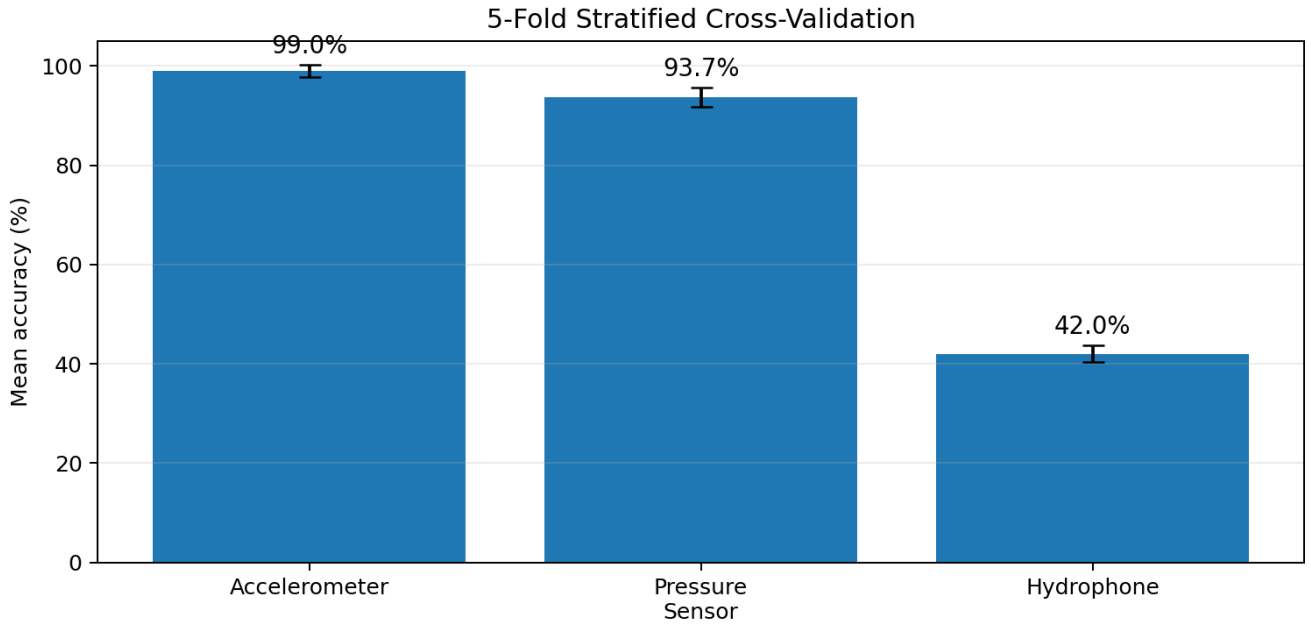
5-Fold Stratified Cross-Validation Protocol

- **Class distribution preserved across folds.**
- StandardScaler fitted strictly on training partition within each fold.
- **Primary Statistical Result: Mean classification accuracy ± standard deviation.**
- Supporting Evidence: Representative test splits (n=102, 109, 180) for per-class metrics and confusion matrices.
- Sampling Rigor: Non-overlapping 1 s segmentation prevents sample sharing between adjacent windows.

Validation Distinction & Protocol Rigor: Primary headline results are the 5-fold stratified CV means (± SD). The 99.2% result is an exploratory grid-search peak on the accelerometer alone. Representative test-split confusion matrices provide class-level error diagnostics; their raw accuracy does not define the headline result.

Comparative Sensor Performance

Primary statistical result: 5-fold stratified cross-validation mean accuracy ± standard deviation



99.0% ± 1.2%
Accelerometer (Non-Invasive)
Macro F1 = 0.9905 · Robustness: High

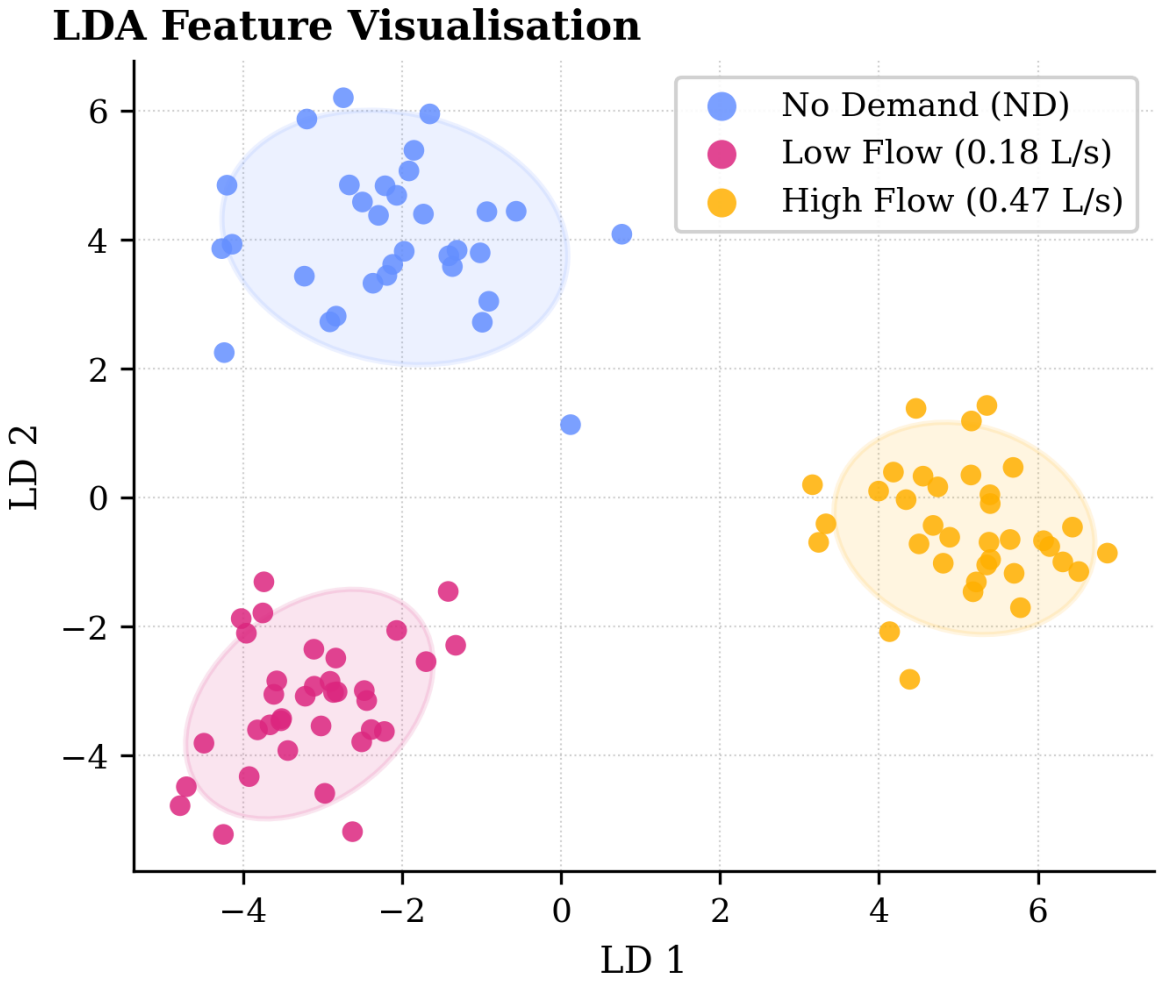
93.7% ± 1.9%
Dynamic Pressure (Invasive)
In-pipe fluid contact · Robustness: Medium

42.0% ± 1.7%
Hydrophone (Invasive)
In-fluid submerged acoustic · Robustness: Low

Benchmark Context & Non-Invasive Finding: Prior work on this benchmark focused on binary leak detection and leak-type classification. Under identical benchmark conditions, non-invasive pipe-wall vibration achieved the highest mean accuracy among evaluated modalities for three-state flow severity estimation.

Accelerometer: Class-Level Evidence Chain

2D LDA feature-space projection & per-class classification metrics (n = 102)



The LDA projection shows clear separation of the three hydraulic-state clusters, supporting the strong accelerometer classification performance. Note: LDA is a feature-space visualization tool, not the final classifier.

Primary Statistical Result

99.0% ± 1.2%

5-Fold Stratified CV Mean · Macro-average F1 = 0.9905

Representative Test Split (Table IV, n = 102)

Class	Precision	Recall	F1	Support
No Demand (ND)	1.0000	1.0000	1.0000	32
Low Flow (0.18 L/s)	1.0000	0.9714	0.9855	35
High Flow (0.47 L/s)	0.9722	1.0000	0.9859	35
Macro Average	**0.9907**	**0.9905**	**0.9905**	**102**

Metric Definitions & Error Analysis:

- **Precision:** Prediction reliability · **Recall:** Detection rate (actual identified)
- **F1-Score:** Harmonic balance 2PR/(P+R) · **Support:** True sample count

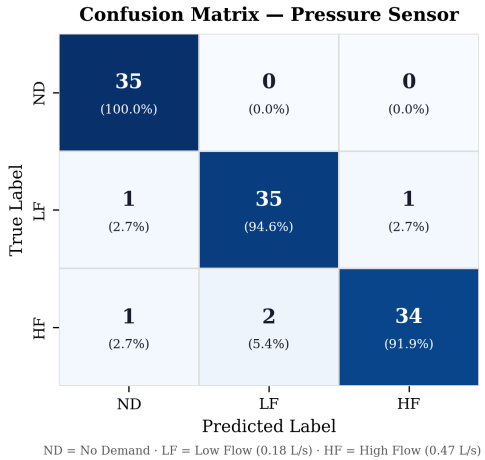
Error Diagnostic: 101 / 102 segments correctly classified. Exactly one sample misclassified: Low Flow → High Flow.

Note: Representative test split shown for class-level/error analysis; headline accuracy is the 5-fold CV mean.

Invasive Sensors: Evidence & Error Analysis

Confusion matrices, compact per-class metrics, and failure mode diagnosis

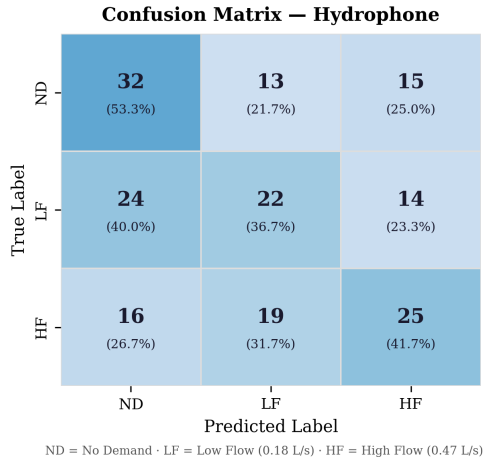
Dynamic Pressure (5-Fold CV: 93.7% ± 1.9%)



Class	P	R	F1	N
No Demand (ND)	.946	1.000	.972	35
Low Flow (0.18 L/s)	.946	.946	.946	37
High Flow (0.47 L/s)	.971	.919	.944	37
Macro Average	**.954**	**.955**	**.954**	**109**

Error Diagnostic (n = 109):
Strong No Demand recognition; residual errors are concentrated around the Low/High boundary, with one additional High → ND error.

Hydrophone (5-Fold CV: 42.0% ± 1.7%)



Class	P	R	F1	N
No Demand (ND)	.444	.533	.485	60
Low Flow (0.18 L/s)	.407	.367	.386	60
High Flow (0.47 L/s)	.463	.417	.439	60
Macro Average	**.438**	**.439**	**.437**	**180**

Error Diagnostic (n = 180):
The paper attributes the poor performance to acoustic saturation/clipping under high-flow turbulence, reducing discriminative spectral structure.

P = Precision | R = Recall | F1 = F1-score | N = Support • *Representative test splits shown for class-level/error analysis; headline accuracy is the 5-fold CV mean.*

Why Do the Sensors Behave Differently?

Physical signal-path interpretation — observed results vs. paper interpretations

Accelerometer (99.0%)

Signal path:
Flow turbulence → pipe-wall structural vibration → external sensor

Why it works:
Vibration amplitude scales with flow rate in the 1–5 kHz structural resonance band of HDPE pipe.

Non-invasive mounting:
External clamp-on placement captures pipe-wall motion directly without fluid contact.

Feature-space evidence:
LDA shows three well-separated clusters — distinct WST signatures per class.

Pressure Sensor (93.7%)

Signal path:
Flow turbulence → fluid pressure fluctuations → invasive transducer

Strong overall performance:
Direct fluid contact provides a high-SNR pressure signal.

Residual confusion:
Low Flow and High Flow states exhibit spectral overlap in the pressure domain at the transition boundary.

Confusion matrix:
Only Low⇔High overlap; No Demand perfectly classified (35/35).

Hydrophone (42.0%)

Signal path:
Flow turbulence → acoustic emission in fluid → in-fluid hydrophone

Physical limitation:
At high flow rates (0.47 L/s) in a 25 mm HDPE pipe, cavitation noise can saturate the acoustic dynamic range, destroying discriminative spectral features.

Confusion pattern:
Widespread confusion across all three classes — consistent with loss of discriminative information due to acoustic saturation.

Interpretation:
In-fluid acoustics in small-diameter pipes under high-turbulence conditions is a challenging modality for severity estimation.

Cross-Sensor Evidence Synthesis

Sensor	CV Accuracy	Signal Domain	Feature Separability	Observed Error Pattern	Paper Interpretation
Accelerometer (Non-Invasive)	**99.0% ± 1.2%**	Structural vibration (1–5 kHz resonance)	Clear 3-class separation in WST feature space	1/102 misclassified (Low → High only)	Flow-magnitude-dependent pipe-wall resonance
Pressure Sensor (Invasive, in-pipe)	**93.7% ± 1.9%**	Fluid pressure transients	Strong ND separation; Low/High boundary overlap	~5/109 errors around Low–High boundary	Network conditions produce pressure overlap
Hydrophone (Invasive, in-fluid)	**42.0% ± 1.7%**	Acoustic emission in fluid column	Broad cross-class overlap in all WST bands	Near-random confusion all three classes	Acoustic clipping/saturation at high flow turbulence

Practical Meaning, Limitations & Future Work

Virtual Flow Meter proxy concept, experimental boundaries, and research roadmap

Practical Relevance & VFM Proxy Concept

Virtual Flow Meter Proxy Scope: External accelerometers as a discrete severity proxy under benchmark conditions. Enables operational triage, not continuous revenue-grade metering.

Non-Invasive Operational Advantage:
Clamp-on placement avoids pipe cutting, eliminates downtime, prevents water contamination.

Complementary SCADA Integration:
Severity classification augments binary alarms — utilities can prioritize 0.47 L/s ruptures over 0.18 L/s leaks.

Theoretical Edge Feasibility:
WST complexity O(N log N); fixed filters + SVM theoretically suitable for low-power edge; empirical device profiling remains future work.

Scientific Limitations

- **Controlled Benchmark Testbed:** Single 25 mm HDPE branched pipe; orifice leaks only.
- **Discrete Severity Levels:** ND, 0.18 L/s, 0.47 L/s only; continuous flow rate not achieved.
- **Laboratory Environment:** No urban traffic, environmental acoustic noise, or soil dampening.
- **Empirical Hardware Profiling:** Device-level execution time and memory footprint on microcontrollers were not tested.

Current Study Scope vs. Future Research Directions

Dimension	Current Study	Future Direction
Measurement scope	Benchmark HDPE testbed (25 mm, branched)	Real municipal distribution networks
Flow severity levels	Discrete: ND / 0.18 L/s / 0.47 L/s	Continuous regression (virtual metering)
Sensor evaluated	External accelerometer (primary winner)	Multi-sensor fusion; optimized modality
Pipe material / topology	HDPE, branched network topology	Metallic, PVC, looped topologies
Feature extraction	WST (J=9, Q=1) + RBF-SVM (C=100, γ=1)	Optimized WST depth; learned representations
Hardware deployment	Theoretical feasibility only (O(N log N))	Empirical MCU benchmarking (cycles, RAM)

Conclusion & Key Takeaways

Synthesis of experimental findings, validation rigor, and operational relevance

99.0% ± 1.2%
Accelerometer
(Non-Invasive)

93.7% ± 1.9%
Dynamic Pressure
(Invasive)

42.0% ± 1.7%
Hydrophone
(Invasive)

The non-invasive accelerometer achieved the highest mean accuracy among the evaluated sensor modalities under the benchmark conditions.

Primary result: 5-fold stratified CV mean ± SD; representative test splits used for class-level error analysis.

Demonstrates that external structural vibration reliably reflects hydraulic severity (ND vs. 0.18 L/s vs. 0.47 L/s) without requiring pipe penetration.

- **Problem:** NRW consumes up to half of supply; utilities require flow severity triage beyond binary alarm.
- **Method:** WST analytical filters (J=9, Q=1) + RBF-SVM (C=100, γ=1) with fold-isolated scaling.
- **Finding:** Non-invasive pipe-wall vibration achieves 99.0% ± 1.2%; invasive fluid sensors 93.7% & 42.0%.
- **Future:** Controlled benchmark baseline establishes feasibility; field network validation is the critical next step.

Thank you